INCH-POUND

MIL-B-9972B (USAF)
04 February 1994
SUPERSEDING
MIL-B-9972A (USAF)
18 Oct 1971

MILITARY SPECIFICATION

BRAZING, NICKEL ALLOY, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification establishes the general requirements for the material, equipment and procedure to be used in the nickel alloy brazing and coating of corrosion and heat resistant alloy parts for operation at temperatures of 1200° Fahrenheit (F) to 2000°F.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

O-A-51	Acetone, Technical
TT-I-735	Isopropyl Alcohol

MILITARY

MIL-E-5007	Engines, Aircraft, Turbojet and Turbofan, General Specifications For
MIL-I-6870	Inspection Requirements, Non Destructive; For Aircraft Materials And Parts
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice, Process For)
MIL-E-8593	Engines, Aircraft, Turboshaft and Turboprop, General Specification For

STANDARDS

FEDERAL

FED-STD-151	Metals, Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Logistics Services Division, SA-ALC/TILDD, Bldg 171, 485 Quentin Roosevelt Rd, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A AREA TIJM

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:17Z
Check the source to verify that this is the current version before use.

MILITARY

MIL -STD-130	Identification Marking of US Military Property
MIL-STD-4 10	Qualification of Inspection Personnel (Magnetic Particle and Penetrant)
MIL-STD-453	Inspection, Radiographic
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection Of

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue., Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Air Force-Navy Aeronautical Bulletin

ANA 343	Specifications and Standards Applicable to Aircraft Engines and Propeller, Use Of

(Copies of specifications, standards, handbooks, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing of brazed and coated parts.

3.2 Preproduction sample. A preproduction sample of parts brazed and coated in accordance with this specification shall be furnished to demonstrate the ability to meet acceptable standards of materials, workmanship, and reliability specified herein. It will also be used to determine compliance with the required functional and dimensional interchangeability specified by the contract, order, or work specification as authorized by the responsible engineering authority.

3.3 Selection of specification and standards. Specifications and standards for necessary materials not specified herein shall be selected according to ANA 343 and MIL-STD-970 except as provided in 3.3.1.

3.3.1 Commercial materials. Commercial materials having suitable properties may be used where on the date of invitation for bids where are no suitable standard materials. These materials shall be approved by the responsible engineering authority.

3.4 Materials. Materials shall be of the highest quality and suitable for the purposes intended. They shall conform to the general requirements specified herein and shall be approved by the responsible engineering authority.

3.4.1 Brazing filler metals. Nickel brazing filler metals having suitable properties may be used as provided in applicable contract, order, or work specification. These alloys shall be heat and corrosion-resistant, and nickel, nickel-chromium, nickel-gold, nickel-gold-chromium, nickel-gold-palladium, or nickel-chromium-silicon alloys. They may be in the form of powder, powder-paste, flux-powder paste, or preformed shapes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:17Z
Check the source to verify that this is the current version before use.

3.4.2 Fluxes. Fluxes may be employed as required to improve wetting and flow of brazing alloy in marginal quality hydrogen or poor inert gas atmospheres where the dew point is too high to be used during vacuum processing. The use of a flux shall be approved by the responsible engineering authority. Fluxes may be in the form of powder, liquid, paste, or flux-powder paste.

3.4.3 Protective atmospheres. Unless otherwise authorized by the responsible engineering authority, pure dry hydrogen or vacuum shall be used as protective atmosphere. Hydrogen, when used, shall be free of sulphur compounds, hydrocarbons and oxygen, and shall have a dew point of not higher than -80°F. Vacuum, when used, shall be maintained at 1 micron or less during actual brazing operation. Other protective atmospheres which may be authorized for brazing and cooling purposes by the responsible engineering authority are argon (dew point not higher than -60°F); endothermic gas (dew point not higher than -60°F); associated ammonia (dew point not higher than +20°F); and purified dried disassociated ammonia (dew point not higher than -100°F). Parts manufacturer from base metals of titanium or other reacting metals shall be processed in vacuum or argon. Parts manufactured of alloys containing titanium may also be brazed in pure dry hydrogen (dew point not higher than -100°F) provided the parts are nickel plated in accordance with TT-I-735 to a thickness of .001 to .002 inch and wiped clean with O-A-51 Acetone or TT-I-735 Isopropyl Alcohol prior to brazing. The protective atmosphere shall prevent decarburization, carburization, scaling, or nitriding, of the material being processed.

3.4.4 Flow arrester. This compound shall be used to prevent excess flow of filler metal during the heating cycle. It may be used in the form of liquid, which may be thinned as required; or powder, which may be mixed with a volatile plastic binder provided in 3.4.5 to provide adhesion. This mixture shall have no harmful effects on the material being brazed.

3.4.5 Cement. The cement shall serve as a volatile plastic binder and vehicle for the powdered brazing filler metal and flow arrester. It shall be mixed with the powdered filler metal to permit application with a brush, glass dropper, spray gun, or applicator. The cement shall retain the filler metal in place until the filler metal has melted, and shall be capable of volatizing at a temperature below its cracking temperature so that it can be discharged from the work area without forming a residue during the brazing cycle. Cement viscosity may be varied to obtain the desired consistency for good brazing and coating.

3.5 Equipment. The equipment shall consist of all apparatus required to prepare the surface for brazing and coating, and perform the repairs specified in applicable work specification, contract, or order. The complete processing equipment shall be subject to inspection by authorized Government Inspectors who shall be given access to the facilities required to determine conformance with this specification.

3.5.1 Furnaces. Furnaces shall be of suitable design and construction for the purpose required, and shall provide uniform temperatures within the working zone. Automatic temperature controlling and recording devices, preferably of the potentiometer type shall be provided to satisfactorily control, and record furnace and work retort temperatures. Temperature variations within the working zones shall not be greater than +/-10° from the control point. Means shall be provided for controlling the furnace atmosphere as required. Vacuum measuring gauges shall be of the emission or alphatron type.

3.5.2 Surface preparation. Suitable surface preparation equipment shall be employed to insure proper cleaning and surface preparation of the part to be brazed and coated.

3.6 Procedure. The nickel alloy brazing and coating procedures shall consist of cleaning, inspection, bright annealing, inspection, applying the braze alloy, processing, inspection, applying the coating alloy, processing, and inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:17Z
Check the source to verify that this is the current version before use.

3.6.1 Cleaning. Parts shall be cleaned to remove deposits of oil, grease, paint, dirt, scale, and other foreign substances that would interfere with the bonding action of the braze and coating materials. A surface roughness of approximately 125 micro inches is desirable, especially for coating, but is not required. If surfaces are prepared by blasting, residual abrasive shall be removed from the surfaces after blasting. Blasting with sand, zirconite or other nonmetallic grit is not recommended. Do not use material that has been previously used to clean aluminum or magnesium, or which is otherwise contaminated. Parts shall be cleaned in accordance with manufacturer's instructions. If necessary, wipe surfaces clean with O-A-51 Acetone or TT-I-735 Isopropyl Alcohol.

3.6.2 Bright annealing. When processing in hydrogen atmosphere, parts shall be bright annealed (precleaned) in pure dry hydrogen, unless otherwise specified in applicable work specification, contract, or order. Upon completion, the part shall be bright, clean, and annealed. The surfaces shall be kept absolutely clean. When feasible, handling shall be accomplished wit gloved hands or by means of suitable holding devices. After surfaces have been prepared, the application of filler metal shall be accomplished as soon afterward as possible to avoid contamination.

3.6.3 Applying the braze and coating filler metal. Braze and coating filler metal provided in 3.4.1 shall be preplaced in such a manner and in such quantity as to produce a satisfactory bond with a minimum amount of run off, globules, and erosion. All tack or spot welding operation, that may be required, shall be performed prior to application of the filler metal.

3.6.4 Processing. Prior to applying the filler metal as provided in 3.6.3, parts shall be reformed to original configuration and cracks shall be straightened and aligned so that both sides of a crack are flush, and the width of the gap does not exceed .010 inch. Patches shall be hand-fitted to provide a clearance of .002 to .008 inch. Spot and tack welding may be used as applicable, or specified to position patches, but such welds shall not be located in areas subjected to high stresses during service. Other means of holding patches may be used as specified in applicable work specification, contract, or order. After the filler metal has been preplaced, as provided in 3.6.3, the parts are placed in a work retort. The retort shall be purged of air, and a positive flow of pure dry hydrogen shall be provided, when processing in hydrogen. The parts are charged into the furnace and held there until the filler metal has melted and formed the desired joint, or bond. The heating rate shall be in accordance with good metallurgical practice, or as specified in contract, order, or specification. After brazing and coating, the parts are cooled in such a manner as to prevent cracking and minimize internal stress, distortion, and decarburization. Cooling from a processing temperature to below the scaling temperature shall be done in a protective atmosphere as provided in 3.4.3.

3.6.4.1 Brazing temperature. The optimum brazing temperature varies with the type and melting temperature of the respective filler metal. Temperature shall be 25°F above actual liquidus and shall not exceed 200°F above maximum liquidus, depending upon the amount of diffusion required; or as specified in applicable contract, order, or specification.

3.6.4.2 Brazing time. Processing time depends upon the time required for the filler metal to melt, as well as the thickness of the parent material. Parts shall be heated until the filler metal melts. After the filler metal melts, heating may be prolonged an additional 15 minutes to aid solution, but washing of a joint shall be avoided. Unless otherwise specified, time shall not exceed 30 minutes at temperature.

3.6.5 Post brazing treatment.

3.6.5.1 Cleaning. After brazing and cooling, residual flux and flow arrester compound shall be removed from the parts by a method not injurious to the parent material, filler metal, or specified surface furnace.

3.6.5.2 Heat treatment of assemblies. After completion of the brazing and coating operation, the parts shall be heat treated as required in accordance with MIL-H-6875 or applicable work specification, contract, or order to provide the required hardness. Heat treating may be accomplished by altering the cooling cycle or as a separate procedure in a tempering furnace.

Source: https://assist.dla.mil -- Downloaded: 202610-17T13:17Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction tests.

4.2.1 Sampling instructions. A minimum of two test parts will be furnished by the procuring activity. The supplier will repair these two parts in accordance with the instructions contained herein, and as may be supplemented by contract or order, in order for the procuring activity to conduct the tests specified in 4.3.1. Each sample will be identified as required and forwarded to the testing location designated by the procuring activity in the letter of authorization. The samples shall be plainly identified by securely attached durable tags marked with the following information: "Sample submitted by (contractor's name) (date) for preproduction tests for application approval in accordance with the requirements of Military Specification MIL-B-9972 under authorization (reference letter of authorization)".

4.2.2 Tests. Preproduction testing shall consist of the following:

4.2.2.1 An inspection shall be performed by the responsible engineering activity prior to testing to ascertain the repair and coating are sound. One set of parts will be tested in accordance with the applicable tests in Groups 200, 300, 500 and 800 of FED-STD-151.

4.2.2.2 Unless otherwise authorized by the responsible engineering authority, one set of preproduction samples shall be subjected to a 150-hour qualification test in accordance with MIL-E-5007 or MIL-E-8593 and subjected to a 300-hour flight test in operational aircraft.

4.2.3 Failure and retest. Failure of any test part to conform to any one of the requirements of this specification shall be cause of disapproval. Failure of any one of test assemblies due to a faulty braze shall be cause for disapproval, unless the failure was due to mishandling or faulty engine operation. Any time a brazed part is disapproved, the contractor may withdraw his product, remedy the deficiency and resubmit his product for additional testing (to be determined at time of resubmission).

4.3 Inspections. Each brazed and coated part shall be subject to inspection and tests as required by the applicable drawings, specifications, contracts, or orders. Inspection shall be conducted by the contractor under the general supervision of the Government Inspector. The inspection requirements specified in MIL-I-6870 shall be performed, unless otherwise specified. If required, certifications shall be accomplished in accordance with MIL-STD-453 and MIL-STD-410.

4.3.1 Visual inspection. All visual examinations shall assure all cracks have been filled with filler metal and the filler metal has been properly fused, and the coating is free of flaws, properly fused, and free of excessive amounts of run-off and globules that present abrupt changes in surface contour or assembly difficulties.

4.3.2 Surface finish. Determination of surface roughness shall be made by physical measurement with a profile tracing element and meter calibrated in micro inches, or by visual comparison of a finished surface with a standards (or sample) specimen having the required surface finish.

5. PREPARATION FOR DELIVERY

5.1 The requirements for preservation and packaging are not applicable to this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:17Z
Check the source to verify that this is the current version before use.

5.2 Marking. The contractor shall identify those parts brazed and coated by his activity. Marking shall be performed in accordance with MIL-STD-130.

6. NOTES

6.1 Intended use. This brazing and coating is intended for repair of cracks and flame eroded areas in gas turbine engine hot section parts, and to provide a corrosion and oxidation resistant coating of nickel alloy. This is especially useful in the repair of parts subject to distortion when weld repaired, or parts that are burned to great extent.

6.2 Provisions for preproduction tests. With respect to services requiring testing, awards will be made only to those contractors whose services have been approved prior to bid opening date.

6.2.1 It is to understood that samples shall be furnished at no cost to the Government, and that the contractors shall pay all transportation charged to and from the point where the tests are made. In the case of failure of the sample or samples submitted, evaluation will be performed by the responsible engineering activity. Consideration will be given to the contractor for additional tests only after it has been shown that changes have been made in the product which the responsible engineering activity considers sufficient to warrant additional tests.

6.2.2 It is to be understood that any nickel alloy brazing and coating supplied under contract shall be identical in every respect to the sample tested and found satisfactory, except for changes approved by the responsible engineering activity. Any unapproved changes from the preproduction sample shall constitute cause for rejection.

6.2.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.3 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

6.4 Key words.

Fluxes
Flow arresters
Furnaces

Custodian
Air Force - 99

Review Activity
Air Force - 11

Preparing Activity:
Air Force - 82

Project THJM-F331

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:17Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-9972B (USAF)	2. DOCUMENT DATE (YYMMDD) 94/02/04

3. DOCUMENT TITLE
BRAZING, NICHEL ALLOY GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME SA-ALC/TILDD	b. TELEPHONE (Include Area Code) (1) Commercial (210) 925-6314	(2) AUTOVON 945-6314
c. ADDRESS (Include Zip Code) 485 Quentin Roosevelt Road Kelly AFB, TX 78241-6425	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-01-17T13:17Z
Check the source to verify that this is the current version before use.